Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/20/2005
PI Name:	Graziosi, David		
Project Title:	Development of a High Performance Mobility/Sizing Actuation System for Application to a Spacesuit Rear Entry Soft Upper Torso and Strength Augmented Mobility Joints		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced extravehicular activity		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	grazid@ilcdover.com	**Fax:**	FY 302-335-4590
PI Organization Type:	INDUSTRY	**Phone:**	302-335-3911
Organization Name:	ILC Dover Inc		
PI Address 1:	One Moonwalker Road		
PI Address 2:			
PI Web Page:			
City:	Frederica	**State:**	DE
Zip Code:	19946-2080	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	06/02/2003	**End Date:**	05/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Akin, David (University of Maryland)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	ILC has teamed with the University of Maryland on the development of a high performance mobility/sizing actuation system. This technology has application in two areas (1) a soft upper torso (SUT) pressure garment for joint interface geometry repositioning to improve specific joint motion, hammering vs. hand over hand translation, etc., and (2) strength-augmented mobility joints The research began with a system requirements analysis to identify and categorize actuation system requirements for a multitude of applications (ISS, RLV, Mars, etc.). System safety and cost parameters, including maintenance issues, will factor heavily into the research. An actuation study wasl be conducted to identify potential actuators that provide acceptable force and response times, and that have limited power requirements. Methods for force multiplication will be researched for application to improve actuation range of motion with reduced power consumption. Methods for applying		

Task Description:	the actuators to reposition the shoulder joint interface angles of the SUT will be investigated to enhance kinematics of the suit assembly for various operations. A SUT will incorporate the mobility/sizing actuation technology in the form of a system of rapidly sizable structural tendons and softgoods that govern interface angles. The actuators will be used to position the SUT joint interface angles in a pre-programmed location and remain there until task completion. The joint interface will then be held in this position until it is again activated. The SUT development will include the creation of an algorithm for the conversion of scan or manual human anthropometric data into optimal SUT shape including optimal interface ring positions. The control mechanism will also be modeled and developed. Attention will be given to developing a fail-safe design that provides redundant function in the event of a loss of power or function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is focusing on force augmented spacesuit pressure garments. The immediate benefits to earth are improved spacesuit technology that will enable future astronauts to explore planetary bodies regardless of the debilitating effects of life in zero gravity. It will provide a means for real-time suit customization by morphing the upper torso to fit each astronaut. This improved fit results in improved comfort, improved mobility, and elimination of fatigue and tissue damage that currently limits the ability for astronauts to perform consecutive EVAs. Advances in anthropometric force augmentation can also have specific benefit in the area of physical rehabilitation. Technologies and systems developed during this research may have application to the medical industry.
Task Progress:	The year 1 deliverables have been completed. These include a requirements study and document, actuator trade study including TRL levels, electronic textile trade study, soft upper torso 3D computer model, and a pressurizable soft upper torso prototype. The soft upper torso prototype was developed as a proof of concept for scye plane 1 degree of freedom linear movement. The goal was to have a manual resizing system on the prototype. The prototype was more advanced in that it was equiped with commercial fluidic muscle actuators. Year 2 is progressing with the development of custom nastic cell actuators. Preliminary testing has been performed on a prototype actuator to characterize its performance. Forward work includes continued development of the nastic cell actuator technology, initial sizing algorithm. initial version of the control model, and demonstration of the integration of the nastic cell actuator technology into a pressurizable soft upper torso prototype.
Bibliography Type:	Description: (Last Updated: 04/13/2017)
Papers from Meeting Proceedings	Graziosi D, Ferl J, Splawn K. "Development of a Space Suit Soft Upper Torso Mobility/Sizing Actuation System." 2004 International Conference on Environmental Systems, Colorado Springs, CO, July 2004. 2004 International Conference on Environmental Systems, Colorado Springs, CO, July 2004. SAE Technical Paper 2004-01-2342, July 2004. http://dx.doi.org/10.4271/2004-01-2342 , Jul-2004
Presentation	David Graziosi "Results to Date on the Development of a High Performance Mobility/Sizing Actuaion System for Application to a Space Suit Soft Upper Torso" Habitation 2004 Jan-2004